

Giovanni Ranieri

CONTACT

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Portfolio: www.flxinxout.github.io

Profile: Master Student at EPFL in Communication Systems. High interest in Mathematics applied to communication, robotics and quantum engineering. *Currently looking for an Internship in industry.*

WORK EXPERIENCE

Vice-President Research – EPFL Xplore | 2025–PRESENT

Lead the EPFL Xplore Research pole — strategy, event programming, and external representation of the association.

Software System Engineer – EPFL Xplore | 2024–2025

Responsible for the entire software system of a Martian-like Rover, leading a team of 20 bachelor and master students in the European Rover Challenge 2025.

Student Assistant – VILAB, EPFL | Fall 2023

Contributed to the creation of the new IC Bachelor course “*Communication System Project*”

Control Station Member – EPFL Xplore | 2023–2024

Control Station web developer and Mikrotik configurations with RouterOS.

EDUCATION

EPFL, Swiss Federal Institute of Technology Lausanne | 2025–PRESENT

Master of Science in Communication Systems

EPFL, Swiss Federal Institute of Technology Lausanne | 2020–2024

Bachelor of Science in Communication Systems – GPA 5/6

Gymnase in Nyon | 2017–2020

High School (Baccalauréat) – Biology and Chemistry option

PROJECTS

Evaluation of NPU Acceleration for Offloading 5G Software Workloads | Spring 2025

FFT Implementation on AMD/Xilinx XDNA/AIE-ML Neural Processing Unit using MLIR-AIE. Evaluation of performance compared to CPU implementation from srsRAN.

Speeding up the Quantification of Data Staleness in DBO | Spring 2024

C++ Implementation of a criterion for a well-suited optimizer for dynamic black-box functions. Evaluation of performance compared to state of the art methods.

- Contribution to NeurIPS’24 Paper “*This too shall pass: Removing Stale Observations in Dynamic Bayesian Optimization*”

Flipper – A Snake Robot | Spring 2023

A 1m20 long Snake Robot designed and built at EPFL (Making Intelligent Things course).

- *Invited to present at annual EPFL IC Section Meeting in Lausanne*
- *Invited to showcase at Viscon2023 in ETH*
- *Interviewed for EPFL News*



SKILLS	Programming & Technologies • C/C++, Java, Python, JavaScript, Assembly, VHDL/Verilog • Docker, Kubernetes, Redis, MongoDB, RabbitMQ, ReactJS, ROS2, AWS	Mathematics & Computer Science • Statistical Signal Processing, Stochastic Models • Digital Communications, Mobile Networks, Satellite Theory • Algorithms, Computer Architecture
	Management • EPFL Xplore (Research & Competition Poles)	Languages • French (Native Speaker) • Italien (Second Language) • English (Fluent)
INTERESTS	Quantum Engineering: Quantum computing and communication	
	Communication Theory: Mathematical Foundations, Mobile Networks, Protocols	
CERTIFICATES	Space industry: Robotics, Applications to Telecommunications	
	European Rover Challenge 2024 & 2025: Twice selected with EPFL Xplore	
OTHER ACTIVITIES	Sport: Climbing, Swimming	
	Students 4 Students: Student Assistant for Analysis and Physics in a preparation week for new EPFL students	